

AI FOR VALIDATION

CHAPTER 5

Language Models for Test-Space Discovery

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When Search Is Not Enough

Chapter 4 searches within the test space; Chapter 5 asks whether the space itself is adequate.

1 CHAPTER 4

Search the admitted space

Mutate, rank and refine candidate tests already allowed by the current representation.

2 CHAPTER 5

Revise the representation

Add a missing factor, interaction, operator or test family when the current space cannot express the right test.

3 KEY QUESTION

Are we looking in the right space?

More optimization cannot find a weakness that the constructor cannot build.

4 TEACHING POINT

Structural discovery is not “search harder.” It is changing what the search is allowed to test.

The model under validation remains fixed; the test representation changes.

Candidate Change vs. Structural Change

The distinction tells us whether ordinary search can continue or the representation must be revised.

1 CANDIDATE-LEVEL CHANGE

Change values inside the current design

Examples:

- Move a threshold
- Swap a medoid
- Change a prompt factor level
- Try a nearby region

The existing constructor can already build the test.

2 STRUCTURAL CHANGE

Change the design itself

Examples:

- Add a new factor
- Add an interaction
- Allow union of regions
- Add a multi-turn conversation structure

The old constructor could not build the test.

3 WHY IT MATTERS

Optimization can refine candidates. It cannot invent an excluded test family unless the representation is changed.

Build the Reflection Context

The LLM receives a bounded evidence packet, not the entire validation repository.

1 WHAT TO INCLUDE

**Current test-space
specification**

**Search history
and candidate definitions**

**Measured diagnostics
and weak-region profiles**

**Outer Array
evaluation conditions**

**Approved domain context
features, units, semantics**

**Known limitations
and representation gaps**

2 WHAT TO EXCLUDE

Keep out of reflection

Reserved confirmation evidence must not be used to craft the proposal.

Otherwise the search contaminates the test that should later confirm it.

3 GOVERNED CONTEXT

Measured facts, interpretations and conjectures must be labeled separately.

The LLM should know what was measured, what is suspected and what the current representation cannot express.

LLM as the Discovery Proposer

The LLM proposes a structural change; it does not score, verify or authorize the finding.

1 ROLE OF THE LLM

Propose a better test space

From the reflection context, suggest:

- a new factor
- an interaction
- a construction rule
- a new test family
- a region combination not currently expressible

2 NOT THE ROLE OF THE LLM

Do not let proposal become proof

The LLM does not:

- assign the score
- validate the evidence
- confirm materiality
- approve the finding
- replace the evaluator

3 SEPARATION OF DUTIES

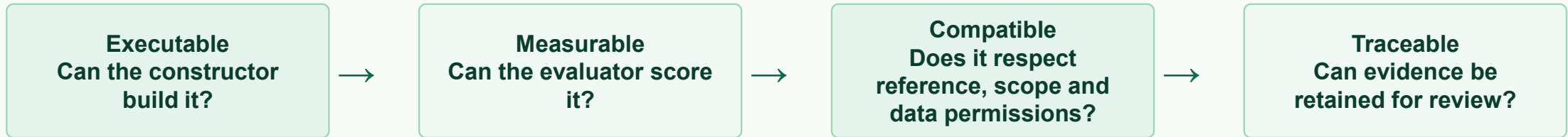
LLM proposes structure. Software constructs and searches. Evaluator measures. Authority decides.

This keeps reflection useful without turning it into “the LLM said so.”

Admission Before Search

A proposed structure must be executable and measurable before it enters the search space.

1 ADMISSION CHECKS



2 AFTER ADMISSION

Return to numerical search

Once admitted, ordinary search can mutate, evaluate and select within the revised test space.

Measure improvement

The revised space must show better discovery evidence than continued search or a baseline.

Keep confirmation separate

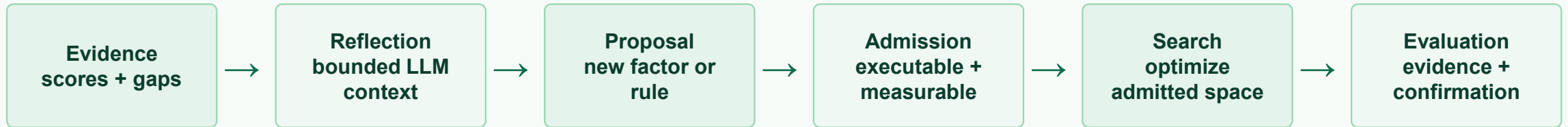
Even a useful discovered structure still needs frozen confirmation on reserved evidence.

An interesting proposal is not yet an admitted test. An admitted test is not yet a confirmed finding.

Proposal → Search → Evidence

The complete Chapter 5 loop: reflection changes the space, then measurement decides whether it helped.

1 STRUCTURAL DISCOVERY LOOP



2 KEY PRINCIPLE

Reflection is useful

The LLM can notice missing structure that ordinary optimization cannot express.

Measurement is decisive

The evaluator determines whether the new structure actually exposes more or different weakness.

Confirmation still governs

Discovered improvements must be frozen and checked before becoming findings.

Chapter 5 adds reflection to the Discovery Loop without letting reflection replace evidence.
